

FarmVerse: Precision Agriculture Management Application

TEAM A

Component Documentation



1. **Current User Profile:** Read

GET /api/users/me

Method: GET,

Endpoint: /farmverse/users/getUserData ? userId = "userId"

Response Payload: (JSON)

```
{  
  "objectId" = "",  
  ...  
}
```

2. **Global Search:** Search/Read

GET /api/search?q={query}

3. **Notifications:** Read/Update

GET /api/notifications

PATCH /api/notifications/{id}/read

4. **Messages / Unread Message Count:** Read

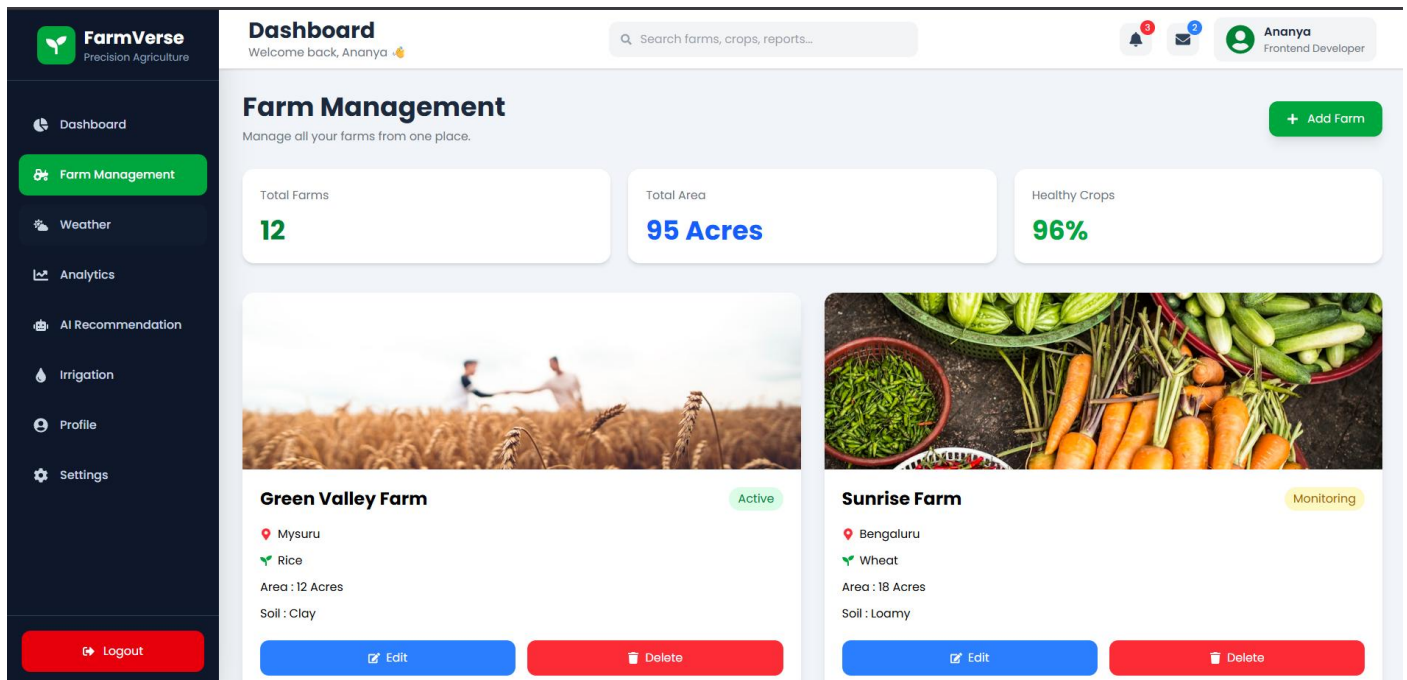
GET /api/messages

GET /api/messages/unread-count

5. **Dashboard Search:** Search/Read

GET /api/search?q={query}

6. **Dashboard Notification Button:** Read/Update
GET /api/notifications
PATCH /api/notifications/{id}/read
7. **Weather Summary:** Read
GET /api/weather/current
8. **Crop Health Summary:** Read
GET /api/dashboard/crop-health
9. **Water Tank Level:** Read
GET /api/dashboard/water-tank
10. **Active Farms Count:** Read
GET /api/dashboard/farms-summary
11. **Crop Yield Analytics:** Read
GET /api/analytics/crop-yield
12. **Detailed Weather Widget:** Read
GET /api/weather/current?location={location}
13. **Logout:** Action
POST /api/auth/logout
14. **AI Insights / Recommended Crop:** Read
GET /api/ai/insights
15. **AI Confidence Score:** Read
GET /api/ai/insights
16. **Fertilizer Recommendation:** Read
GET /api/ai/recommendations/fertilizer
17. **Recent Activities:** Read
GET /api/activities/recent
18. **Farm Details Summary:** Read
GET /api/dashboard/farm-summary
19. **Crop Status List:** Read
GET /api/crops/status
20. **View All Crop Status:** Read
GET /api/crops
21. **Crop Health and Moisture Data:** Read
GET /api/crops/{cropId}/status
22. **Upcoming Tasks:** Read
GET /api/tasks/upcoming
23. **Farm Progress:** Read
GET /api/farms/progress



Farm Management Page

1. **Farm Summary Metrics:** Read
GET /api/farms/summary
2. **Farm List:** Read
GET /api/farms
3. **Add Farm:** Create
POST /api/farms

Method: POST

Endpoint: /farmverse/farms/addFarms

Request Payload: JSON

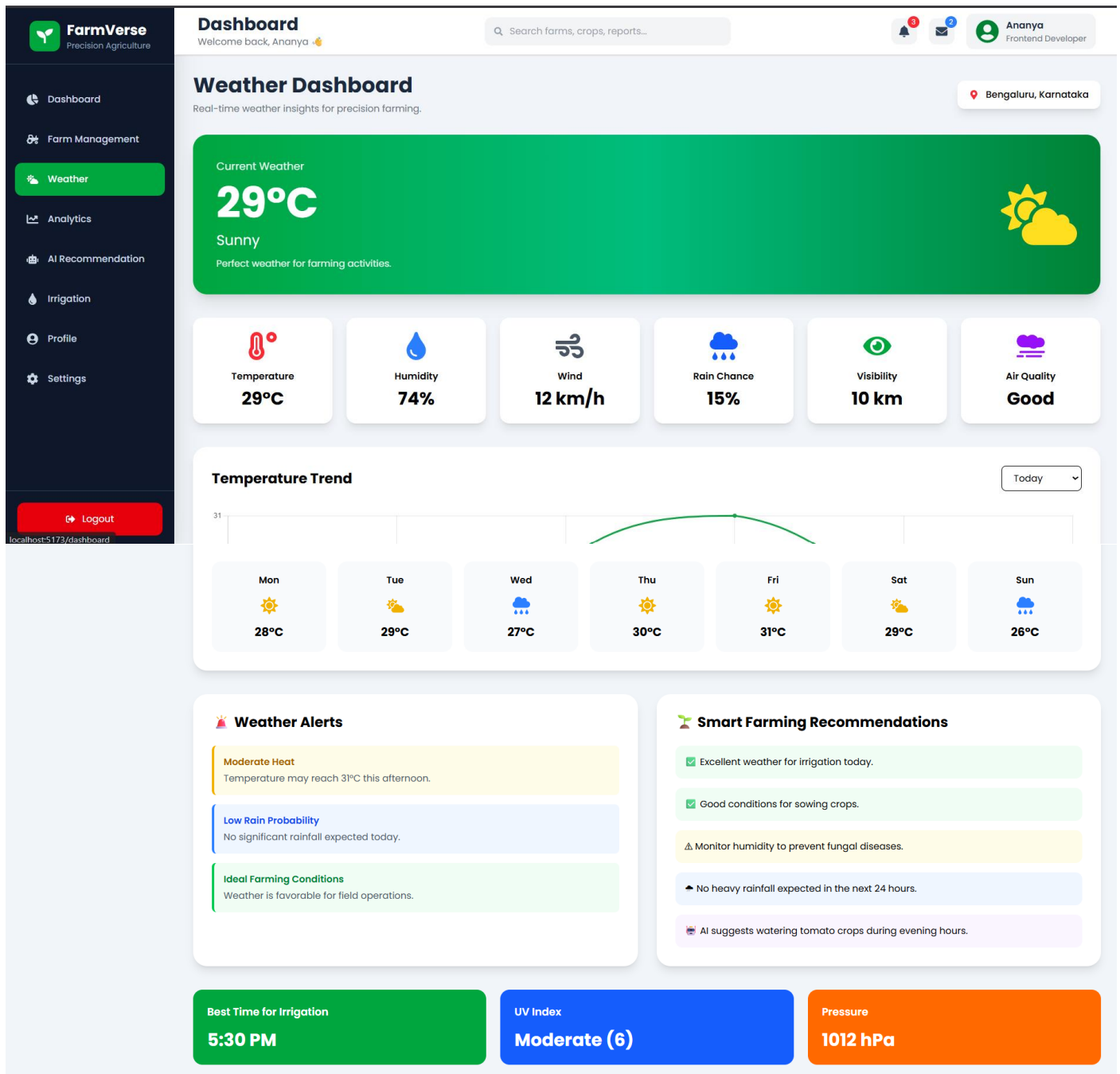
```
{  
  
  "farmName": "",  
  
  ...  
}
```

Response Payload

```
{  
  
  "status": "ok",  
  
  "statusCode": "200",  
  
  "message": "<should cite error/response (eg: 'Farm added successfully')>",  
  
  "id": "<id of object created>"  
}
```

(Maintain logs for changes taking place inside database)

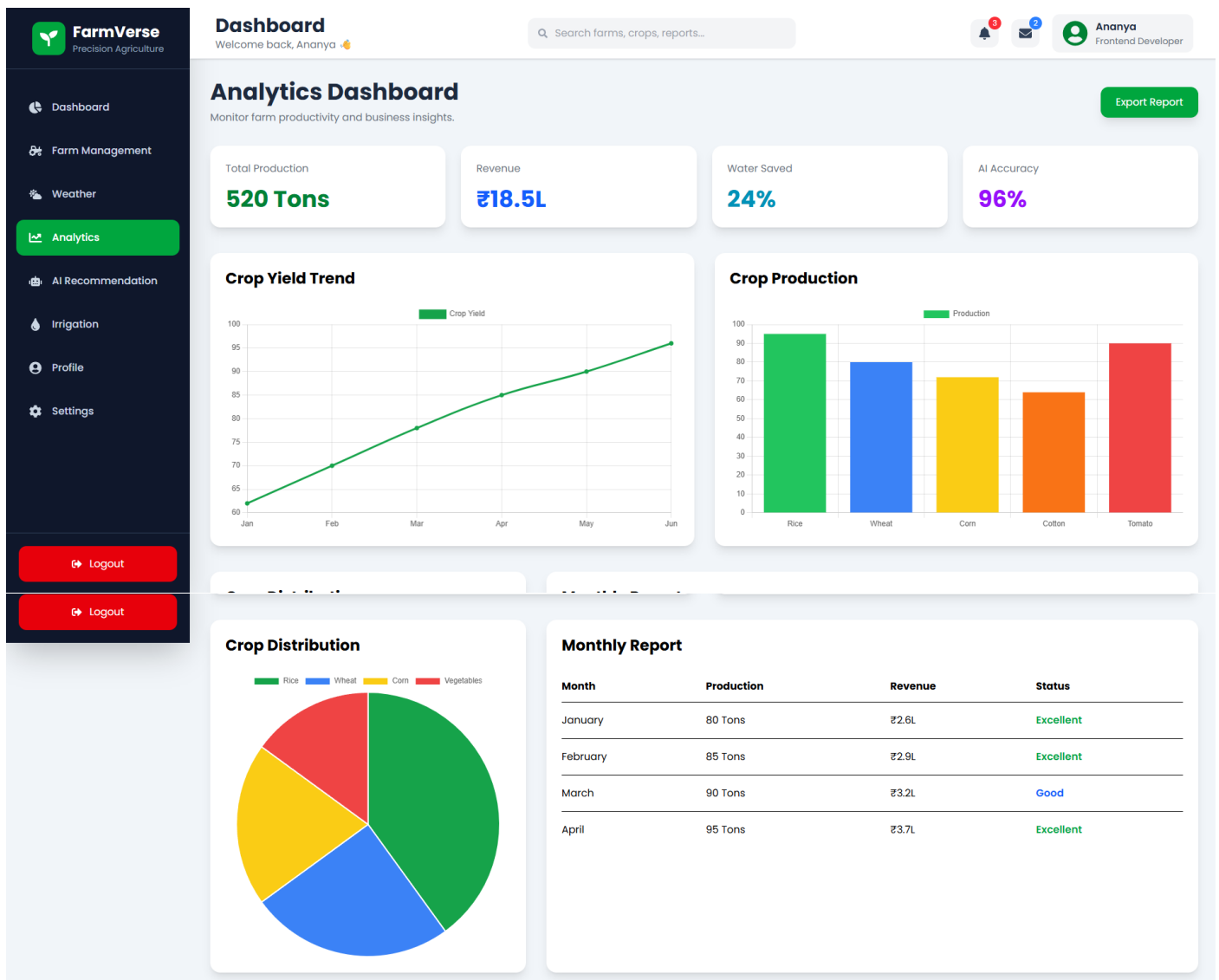
4. **Farm Details:** Read
GET /api/farms/{farmId}
5. **Edit Farm:** Read/Update
GET /api/farms/{farmId}
PUT /api/farms/{farmId}
6. **Delete Farm:** Delete
DELETE /api/farms/{farmId}
7. **Farm Status:** Read
GET /api/farms/{farmId}/status



Weather Page

1. **Current Weather:** Read
GET /api/weather/current

2. **Weather Location:** Read/Update
GET /api/weather/location
PUT /api/weather/location
3. **Weather Metrics (Temperature, Humidity, Wind, Rain Chance, Visibility, Air Quality):** Read
GET /api/weather/current
4. **Temperature Trend:** Read
GET /api/weather/temperature-trend?period={period}
5. **Temperature Trend Period Filter:** Read
GET /api/weather/temperature-trend?period={period}
6. **Weekly Weather Forecast:** Read
GET /api/weather/forecast
7. **Weather Alerts:** Read
GET /api/weather/alerts
8. **Smart Farming Recommendations:** Read
GET /api/weather/farming-recommendations
9. **Best Time for Irrigation:** Read
GET /api/weather/irrigation-time
10. **UV Index:** Read
GET /api/weather/uv-index
11. **Atmospheric Pressure:** Read
GET /api/weather/pressure



Analytics Page

1. **Analytics Summary Metrics:** Read
GET /api/analytics/summary
2. **Crop Yield Trend:** Read
GET /api/analytics/crop-yield-trend
3. **Crop Production Analytics:** Read
GET /api/analytics/crop-production
4. **Crop Distribution:** Read
GET /api/analytics/crop-distribution
5. **Monthly Report:** Read
GET /api/analytics/monthly-report
6. **Export Report:** Download/Export
GET /api/analytics/export-report

**FarmVerse**
Precision Agriculture

Dashboard

Farm Management

Weather

Analytics

AI Recommendation

Irrigation

Profile

Settings

Logout

Logout

Dashboard
Welcome back, Ananya 🌻

🔍 Search farms, crops, reports...

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Ananya
Frontend Developer

AI Crop Recommendation
Smart AI-powered recommendations for better farming decisions.

Generate Recommendation

Enter Farm Details

Farm Location

Select Soil Type

Temperature (°C)

Humidity (%)

Rainfall (mm)

Soil pH

Recommended Crop
Rice

AI Confidence
97%

Expected Yield
High

Water Requirement
Medium

**AI Analysis**
Why AI selected this crop

**AI Analysis**
Why AI selected this crop

✓ Soil nutrients are ideal for cultivating Rice.

✓ Current rainfall prediction supports healthy growth.

✓ Temperature and humidity are within the recommended range.

✓ AI predicts approximately 95–98% yield with proper irrigation.

Farm Information

Bengaluru

Soil : Loamy

Rainfall : 120 mm

Temperature : 29°C

Humidity : 74%

Future AI Features

✓ Fertilizer Recommendation

✓ Disease Detection

✓ Pest Prediction

✓ Harvest Prediction

✓ Profit Estimation

AI Crop Recommendation Page

- Soil Type Options:** Read
GET /api/soil-types
- Generate Crop Recommendation:** Create/Action
POST /api/ai/crop-recommendation
- Recommended Crop Result:** Read
GET /api/ai/crop-recommendation/{recommendationId}

4. **AI Confidence, Expected Yield and Water Requirement:** Read
GET /api/ai/crop-recommendation/{recommendationId}
5. **AI Analysis / Recommendation Reasons:** Read
GET /api/ai/crop-recommendation/{recommendationId}/analysis
6. **Farm Information:** Read
GET /api/farms/{farmId}


FarmVerse
Precision Agriculture

Dashboard

Welcome back, Ananya 🌱





Ananya
Frontend Developer

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Smart Irrigation

Monitor and control irrigation efficiently.



Soil Moisture

74%



Water Tank

82%



Pump Status

ON



Next Schedule

6:00 AM

Pump Controls

Start Pump

Stop Pump

Irrigation History

Date	Time	Duration	Status
07 Jul	06:00 AM	30 mins	Completed
06 Jul	06:00 AM	25 mins	Completed

Irrigation History

Date	Time	Duration	Status
07 Jul	06:00 AM	30 mins	Completed
06 Jul	06:00 AM	25 mins	Completed
05 Jul	06:00 AM	28 mins	Scheduled

AI Irrigation Suggestion


AI Irrigation Suggestion


Soil moisture is sufficient for the next 12 hours.

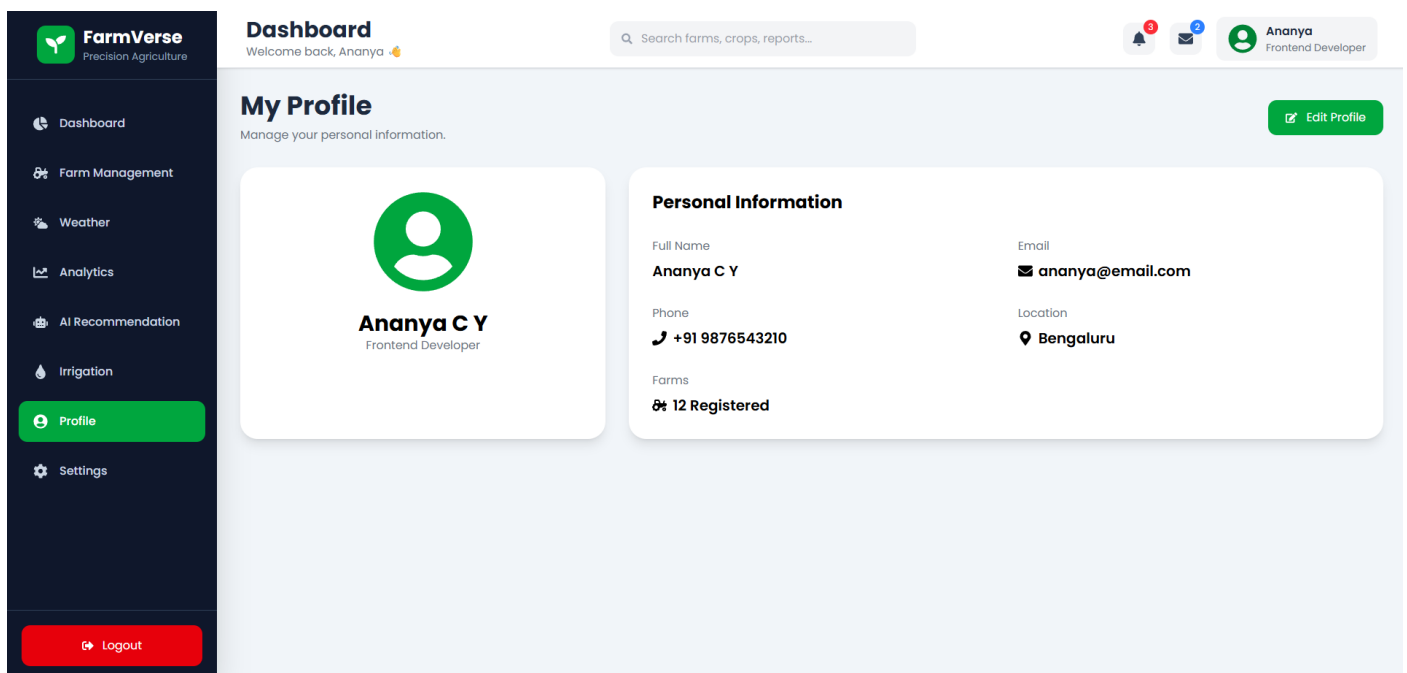

Irrigation is recommended tomorrow morning.


Estimated water saving: 18%.

Smart Irrigation Page

1. **Irrigation Summary / Sensor Metrics:** Read
GET /api/irrigation/summary
2. **Soil Moisture Level:** Read
GET /api/irrigation/soil-moisture
3. **Water Tank Level:** Read
GET /api/irrigation/water-tank

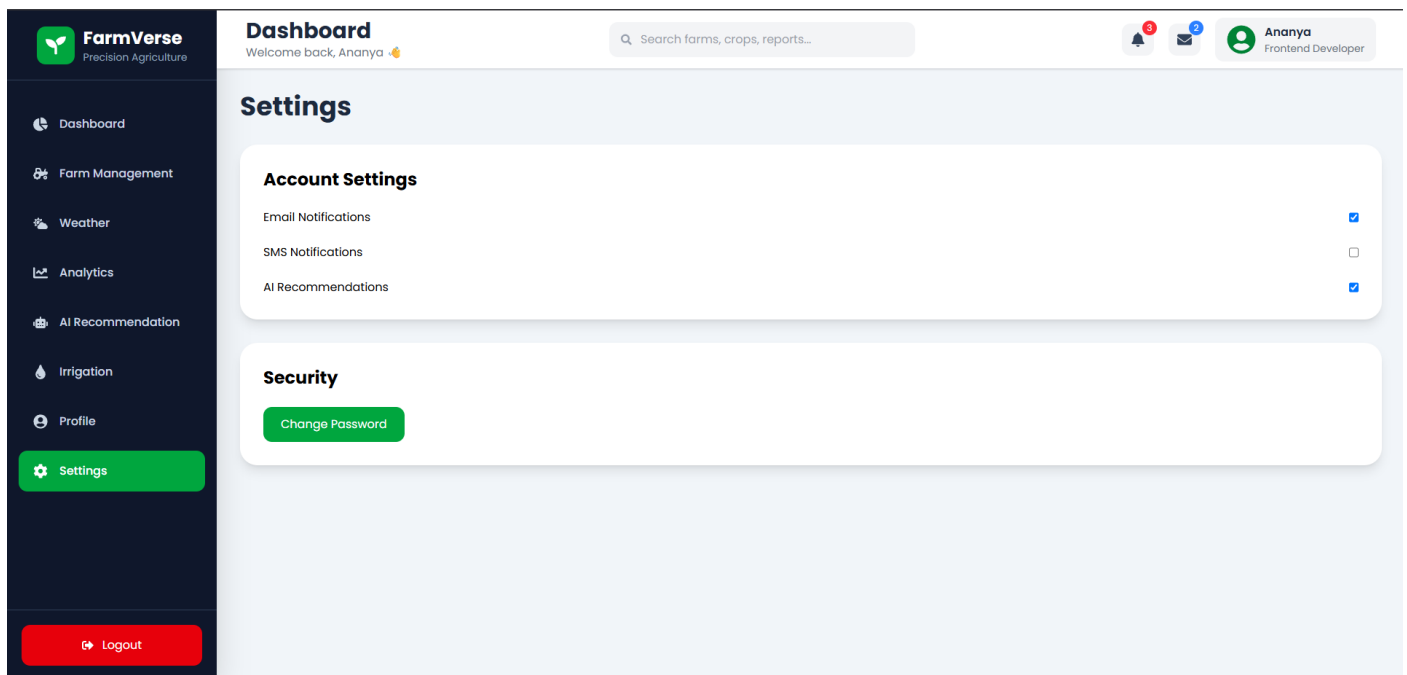
4. **Pump Status:** Read
GET /api/irrigation/pump/status
5. **Next Irrigation Schedule:** Read
GET /api/irrigation/schedules/next
6. **Schedule Irrigation:** Create
POST /api/irrigation/schedules
7. **Start Pump:** Action/Update
POST /api/irrigation/pump/start
8. **Stop Pump:** Action/Update
POST /api/irrigation/pump/stop
9. **Irrigation History:** Read
GET /api/irrigation/history
10. **AI Irrigation Suggestions:** Read
GET /api/ai/irrigation-suggestions



Profile Page

1. **User Profile Information:** Read
GET /api/users/me
2. **Registered Farms Count:** Read
GET /api/farms/summary
3. **Edit Profile:** Update
PUT /api/users/me

4. **Profile Avatar: Read/Update**
GET /api/users/me/avatar
PUT /api/users/me/avatar



Settings Page

1. **Account Settings / Notification Preferences: Read**
GET /api/settings/preferences
2. **Email Notifications: Update**
PATCH /api/settings/preferences
3. **SMS Notifications: Update**
PATCH /api/settings/preferences
4. **AI Recommendations Preference: Update**
PATCH /api/settings/preferences
5. **Change Password: Update**
PUT /api/users/me/password